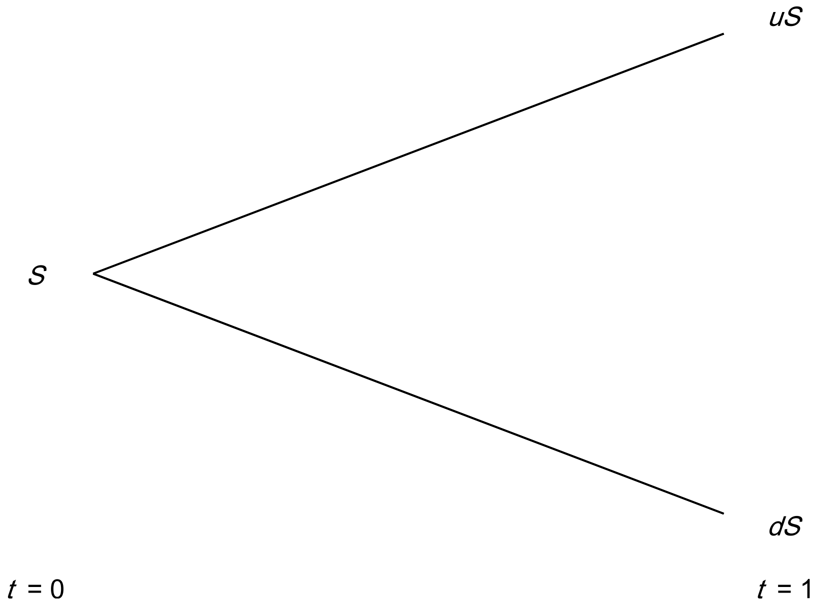

Figure A.3: One-Period Binomial Tree

these probabilities whatsoever.[19] More important, this means that in order to price options, we don't have to worry about trying to figure out what these probabilities are.

There was nothing special about this particular numerical example. We can, and will, do this in general. Consider an asset whose current price is S and will be either uS or dS at the end of one period, where $u > 1$ and $d < 1$ and $d < 1 + r < u$, where r is the risk-free rate over the period assuming simple interest. The evolution of the asset price is shown in Figure A.3. Why do we need to require that $d < 1+r < u$?[20] Denote the up and down payoffs of the call by c_u and c_d, as depicted in Figure A.4.

We wish to set up a hedge portfolio with parameters h and B such that

$$\begin{aligned} huS - BR &= c_u \\ hdS - BR &= c_d, \end{aligned} \tag{A.11}$$

[19]It is true, however, that changing these probabilities may change today's asset price, which would affect the value of the call. But taking today's asset price as given, the up and down probabilities don't matter at all.

[20]If $1 + r < d < u$, then there would be an arbitrage by buying the risky asset and selling bonds. And if $d < u < 1 + r$, there would be an arbitrage by buying bonds and selling the risky asset.

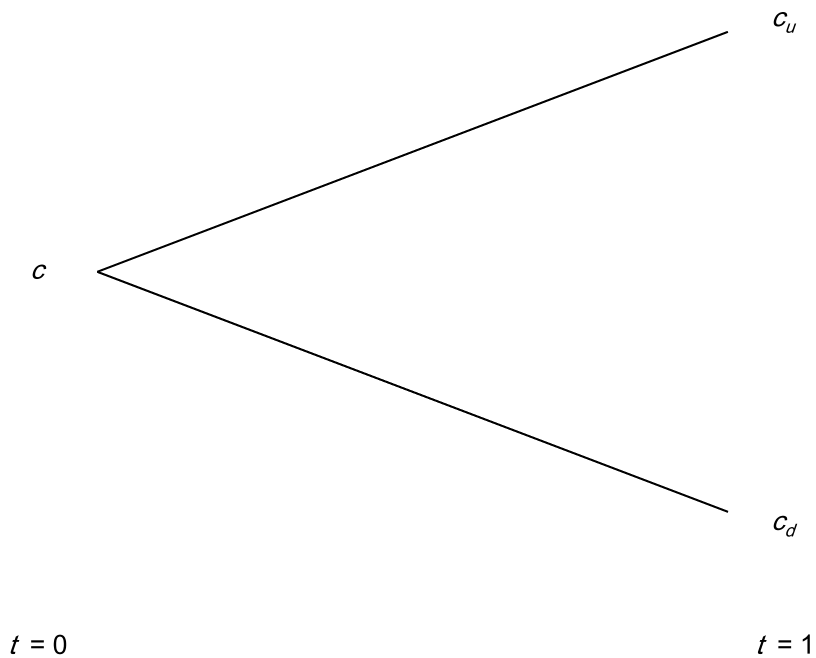

Figure A.4: Payoffs from Call Option

where $R = 1 + r$. We can solve for h and B in terms of the other given variables and find

$$h = \frac{c_u - c_d}{(u-d)S} \tag{A.12}$$

$$B = \frac{dc_u - uc_d}{(u-d)R}. \tag{A.13}$$

By the assumed lack of arbitrage, the value of the call must equal the value of the hedge portfolio today:

$$c = hS - B, \tag{A.14}$$

where h and B are as given in Equations A.12 and A.13, respectively.[21]

[21] The value of the hedge portfolio today is the amount of cash we have to take out of our pocket to set up the hedge portfolio. We are buying h shares of the underlying at a cost of S per share, for a total cost of hS. However, we are simultaneously borrowing B today to partially finance this purchase, which means our total cash outlay is $hS - B$.

Equation A.14 is the value of the call option that we derived under an assumed lack of arbitrage. A few lines of algebra show that the expression in Equation A.14 can be rewritten as

$$c = \frac{1}{1+r}\left[p_n c_u + (1-p_n)c_d\right], \tag{A.15}$$

where

$$p_n = \frac{R-d}{u-d}. \tag{A.16}$$

Equation A.15 is completely equivalent to Equation A.14, but the appeal of Equation A.15 is that it looks as though we are computing the expected payoff of the call and discounting back at the risk-free rate. This valuation expression is intuitively pleasing and has come to be known as *risk-neutral valuation*.[22] The "probabilities" p_n and $1-p_n$ do indeed have the properties that they are between 0 and 1, and of course they sum to 1, but they are not really probabilities. They are, however, often referred to as *risk-neutral probabilities*. Risk-neutrality is an artifical construct but one that makes option pricing intuitive. The risk-neutral expression in Equation A.15 may sometimes be denoted by notation like

$$c = \frac{1}{1+r}\left[\hat{E}(c_T)\right], \tag{A.17}$$

where the " ˆ " denotes that expectations are taken using the risk-neutral probabilities p_n and $1-p_n$. We can appeal to risk-neutrality to price up the call in the numerical example we considered earlier. In this case, we have

$$p_n = \frac{4/3 - 0.5}{1.5 - 0.5} = 5/6,$$

and the value of the call is

$$c = \frac{1}{1.\overline{33}}\left[\$8 \times (5/6) + \$0 \times (1/6)\right],$$

which is equal to \$5, just as we found above.

Let's go back to Equation A.14, the value of the call option obtained by setting up a hedge portfolio. It is easy to show from Equations A.12 and A.13 that $0 \leq h \leq 1$ and $B \geq 0$, which shows that a call option can be thought of as a levered position in the underlying asset. That is, a call option is equal to

[22] The term *risk-neutral* comes from the fact that someone who is risk-neutral is indifferent between a random outcome and its expected payoff. If someone were risk-neutral, they would value a risky asset by computing its expected payoff in the future and discounting at the risk-free rate.

a (fractional) long position in the underlying asset that is partially financed by borrowing. Similarly, a put option can be thought of as a (fractional) short position in the underlying asset, the proceeds of which are lent out.[23]

Also note that in both Equations A.14 and A.15 there was nothing particularly important about the fact that the options valued were calls. The same approach can be taken to value any derivative, v, that has payoffs of v_u in the upper state and v_d in the lower state. In this case, appealing to risk-neutrality gives the value of the derivative as

$$\frac{1}{1+r}\left[p_n v_u + (1-p_n)v_d\right], \tag{A.18}$$

where p_n is as specified in Equation A.16.

A.4.1 Multiperiod Extensions

The binomial pricing model generalizes to multiperiods. As a concrete example, consider the following two-period model in which an asset's possible prices are set out in Figure A.5. Assume again that the risk-free rate is $33\frac{1}{3}\%$ each period. What is the price today of a European call option on the asset with a strike of \$60 that expires in two periods at $t = 2$? The way to solve for the price of the call is to work "backwards" in the tree. At time $t = 1$, we will either be at the upper node or the lower node. If we are at the upper node, we know that one period later at $t = 2$ the asset will be worth either \$72 or \$24. We can use our one-period binomial methodology to find the value of the call at the upper node at time $t = 1$. Using Equation A.15, it can be verified that the value is $(3/4) \times [(5/6) \times 12 + (1/6) \times 0] = \$7.5$. Similarly, if we are at the lower node at time $t = 1$, the call will be worth \$0 at time $t = 2$, regardless of which outcome subsequently occurs. Thus, if we find ourselves at the lower node at time $t = 1$, the value of the call is surely \$0, which we can also verify using Equation A.15 if there is any doubt in our minds whatsoever. Moving back in the tree to time $t = 0$, we can find the value today of the call that will be worth either \$7.5 in the upper node or \$0 in the lower node at time $t = 1$. The value of the call today is $(3/4) \times [(5/6) \times 7.5 + (1/6) \times 0] = \$4.6875$. Figure A.6 lays this out. The

[23]Given these interpretations, we can now go back to answer (at least from the perspective of the binomial model) one of the questions posed in Section A.1: why do calls increase in value as the risk-free rate increases? A call is equivalent to a levered position in the underlying financed by borrowing. As the risk-free rate increases, the cost of borrowing goes up. That is, the cost of the equivalent levered position goes up—that is, it is harder to finance—and therefore the call is worth more. A similar argument shows that puts decline in value as the risk-free rate increases.

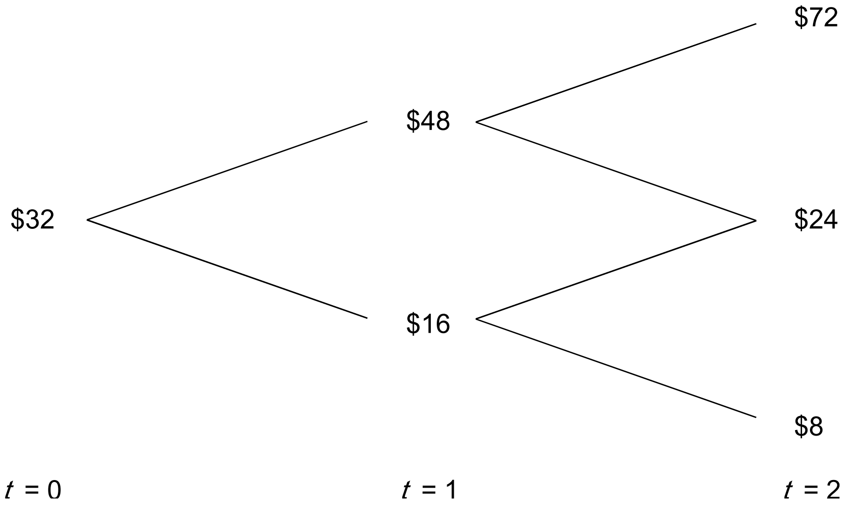

Figure A.5: Two-Period Binomial Tree

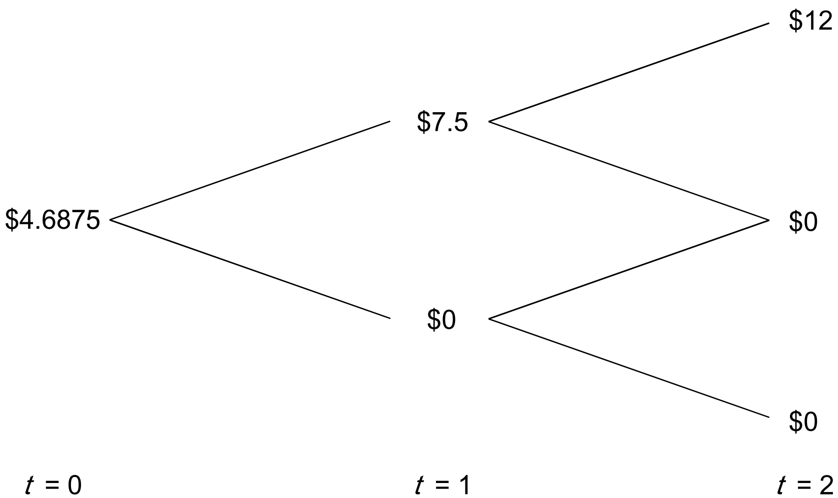

Figure A.6: Valuing the Call Going Backwards approach demonstrated in this simple two-period example easily generalizes to multiperiods of any dimension. See Hull (2008) for more details.

A.5 The Black-Scholes Formula

A multiperiod binomial model is a powerful paradigm used to price a variety of options. It is a *discrete* model, meaning that the underlying asset price moves discretely, once per period. In comparison, the Black-Scholes model considers a continuous distribution, in which the asset price can move at any instant in time.

In particular the Black-Scholes model assumes that the underlying asset price is lognormally distributed. We say that a random variable Y is lognormally distributed with parameters μ and σ^2 if $Y = e^X$, where X is normally distributed with mean μ and variance σ^2. The variable X is said to be normally distributed with mean μ and variance σ^2, denoted $X \sim N(\mu, \sigma^2)$, if the *probability density function* for X is given by

$$n(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}. \tag{A.19}$$

The *cumulative distribution function* for $X \sim N(\mu, \sigma^2)$ is given by

$$N(d) = \int_{-\infty}^{d} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} dx. \tag{A.20}$$

It can be shown that $E(X) = \mu$ and $\text{var}(X) = \sigma^2$. For the case in which $\mu = 0$ and $\sigma^2 = 1$, we say that X is a *standard normal* random variable. Note that if Z is a standard normal random variable, then the random variable $X = aZ + b \sim N(b, a^2)$.

A random variable Y that is lognormally distributed with parameters μ and σ^2 can be shown to have a probability density function given by

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma y} e^{-(\ln(y)-\mu)^2/2\sigma^2}, \text{ for } y > 0. \tag{A.21}$$

It can be shown that

$$E(Y) = e^{\mu+\sigma^2/2} \tag{A.22}$$

and

$$\text{var}(Y) = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1). \tag{A.23}$$

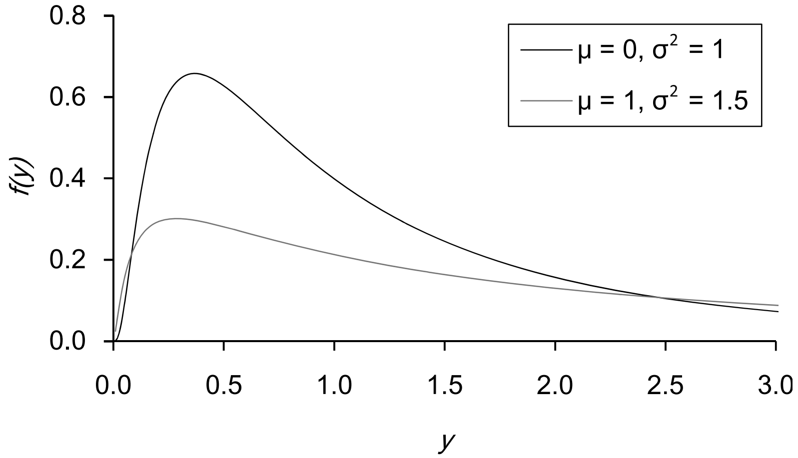

Figure A.7: The Lognormal Density Function

Note that the random variable Y can take on any positive value, whereas the variable X can take on any value. A picture of the probability density function of various lognormally distributed random variables is shown in Figure A.7.

Black-Scholes assumes that the underlying asset price at time T is lognormally distributed. In particular, let's assume

$$S_T = Se^{\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z}, \tag{A.24}$$

where Z is a standard normal random variable, and S is today's asset price.

Note that S_T as defined in Equation A.24 is indeed lognormally distributed, since the exponent in the equation is a normal random variable. If S_T is lognormally distributed as in Equation A.24, then using Equation A.22 and A.23 it can be shown that

$$E[S_T] = Se^{\mu T} \tag{A.25}$$

and

$$\mathrm{Var}[S_T] = S^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right). \tag{A.26}$$

From Equation A.24, it follows that

$$\ln(S_T/S) \sim N\left[\left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right]. \tag{A.27}$$

Thus, $\ln(S_T/S)$ is normally distributed with mean $(\mu - \sigma^2/2)T$ and variance $\sigma^2 T$. Lognormal distributions are pretty reasonable models of the distribution of many asset prices, like equities. As we discussed in Chapter 5, lognormality is often a questionable assumption for the distribution of interest rates.

Black and Scholes (1973) showed that if the underlying asset price is lognormally distributed and pays no dividends, then the value of a call option on the underlying asset struck at X that matures at time T is given by

$$c = SN(d_1) - Xe^{-rT}N(d_2), \tag{A.28}$$

where

$$d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{A.29}$$

$$d_2 = d_1 - \sigma\sqrt{T}, \tag{A.30}$$

where $N(\cdot)$ is the cumulative normal distribution function as given in Equation A.20. Equation A.28 is known as the *Black-Scholes formula*. One of the (somewhat remarkable) things to notice about the Black-Scholes formula is that it does not include the parameter μ despite it appearing in the specification of the underlying asset price dynamics as specified in Equation A.24. The beauty of this is that we do not need to know the value of the parameter μ in order to price the option.

We are not going to concern ourselves with the actual derivation of the Black-Scholes formula. There are a number of ways to derive it. In their original paper, Black and Scholes (1973) took an approach somewhat analogous (although significantly more complicated mathematically) to setting up a hedge portfolio as we did with the binomial model.[24] They showed that such restrictions and the assumed lack of arbitrage dictate that the price of the call is as in Equation A.28.

Since that time, others have shown that the Black-Scholes formula can be priced under the assumption of risk-neutrality. That is, it can be shown that the value of the call as in Equation A.28 is consistent with assuming that the underlying asset price is lognormally distributed as per Equation A.24 and is the solution to computing the following expectation:

$$c = e^{-rT}\hat{E}[\max(S_T - X, 0)], \tag{A.31}$$

[24] Actually, if you want to get technical, they set up a portfolio consisting of a position in the call option and a position in stock and derived conditions on the portfolio such that it was risk-free. These conditions placed restrictions on the call price, the solution to which was shown to be the Black-Scholes formula.

where $\hat{E}$ means that expectations are taken assuming risk-neutrality. So what does that mean? This amounts to replacing the "μ" by an "r" in Equation A.27:

$$\ln(S_T/S) \sim N\left[\left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right]. \tag{A.32}$$

This is pretty important. The parameter μ is asset specific. That is, each underlying asset has its own expected return. How would you go about figuring out what this is? Fortunately, you don't have to. Under risk-neutrality, all assets can be valued assuming they earn an expected return equal to the risk-free rate.

Using put-call parity, it is easy to show using Equation A.28 that according to Black-Scholes the value of a European put on the underlying asset is

$$p = Xe^{-rT}N(-d_2) - SN(-d_1),$$

with d_1 and d_2 as defined in Equations A.29 and A.30, respectively. Figures A.8 and A.9 show the Black-Scholes value of a call option and a put option as a function of the underlying asset price.

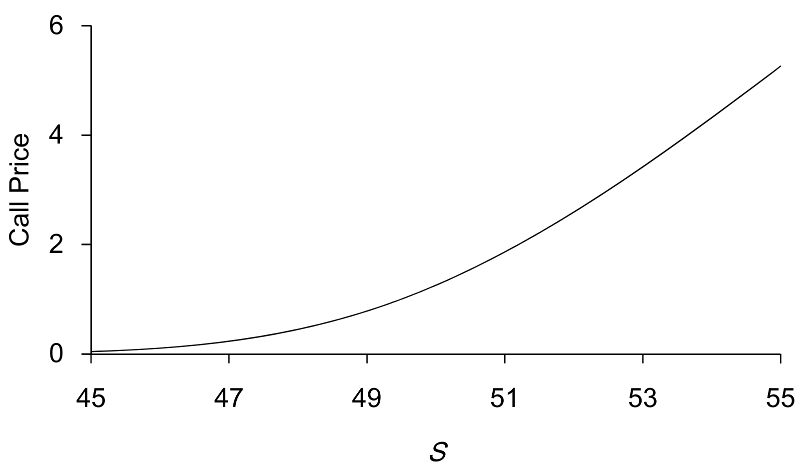

Figure A.8: Black-Scholes Call Option as a Function of Underlying Asset When $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

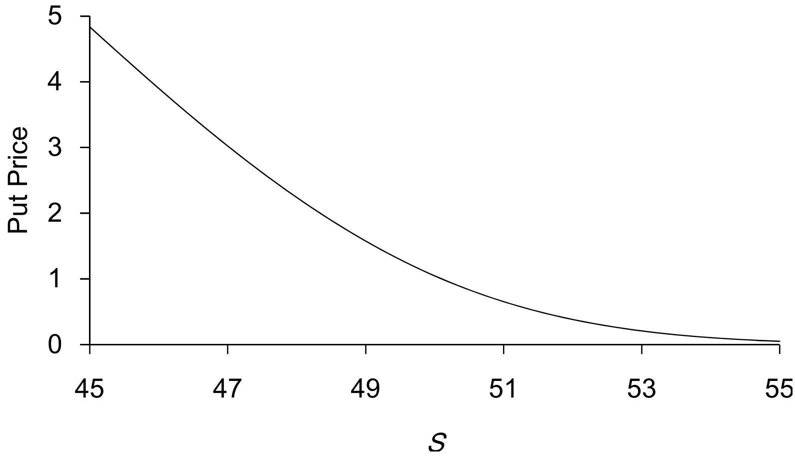

Figure A.9: Black-Scholes Put Option as a Function of Underlying Asset When $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

Black-Scholes with Dividends

If the underlying asset pays a continuous dividend yield d, the value of a European call option on the asset according to Black-Scholes is

$$c = Se^{-dT}N(d_1) - Xe^{-rT}N(d_2), \tag{A.33}$$

where

$$d_1 = \frac{\ln(S/X) + (r - d + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{A.34}$$

$$d_2 = d_1 - \sigma\sqrt{T}. \tag{A.35}$$

Black's Formula

An application of the Black-Scholes formula on a dividend-paying asset in Equation A.33 can be used to value a European call option on a forward contract or a futures contract. Black (1976) showed that the value of a European call option on a forward is given by

$$c = e^{-rT}\left[FN(d_1) - XN(d_2)\right], \tag{A.36}$$

where

$$d_1 = \frac{\ln(F/X) + \sigma^2 T/2}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T},$$